EE4309 Robot Perception

Topic 1: Introduction

Asst Prof. Mike Shou

Instructor

- Lecturer: Mike Shou

Current: 2021 - present, Assistant Professor at NUS
Email: mikeshou@nus.edu.sg
Show Lab: https://sites.google.com/view/showlab

Research Interests: Computer Vision, Multimedia, AR/VR

Awards:

- 2024, NUS Outstanding Early-Career Award
- 2024, ST Engineering Distinguished Professor
- 2022/23, EgoVis Distinguished Paper Award
- 2022, Forbes 30 under 30 Asia list
- 2022, CVPR best paper finalist.
- 2021, Singapore National Research Foundation Fellowship.

Multimodal Video Understanding

NOTE: 1. Audio is supported by ChatTTS for demonstration purpose. Other aspects is supported by our end-to-end streaming video large language model.
2. Pauses in the video is for waiting the slow audio. The inference speed of the model can approach real-time (5~10 FPS on RTX 3090 GPU, 10~15 FPS on A100 GPU).

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Multimodal Video Generation

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Figure 01 = OpenAI's ChatGPT + Robot

Instructor

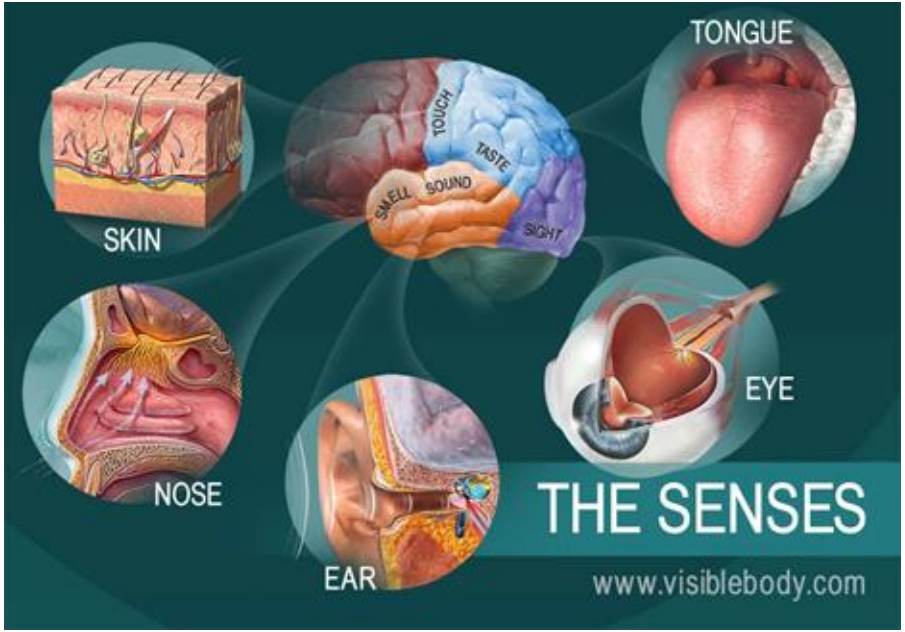
- Graduate Teaching Assistant

1. Yanzhe Chen chenyanzhe@u.nus.edu
2. Guian Fang guianfang@u.nus.edu

Multimodal

The human five senses:

- Sight (Vision)
- Hearing (Auditory)
- Smell (Olfactory)
- Taste (Gustatory)
- Touch (Tactile)



Input from sensors

Brain as processing unit

Output/Determine actions to interact with the world

Vision is probably the most important sense for us to perceive the world

- Recognize person
- Understand scene
- Locate objects
- ……

Audio is also quite useful

Sound source localization

Language

- Further, audio not just about sound - spoken language which makes communication much easier
- Of course, it could be in other textual formats like sign and writing.

Human → Robot

We want to power robot with similar senses:

- Vision
- Audio
- Language
- …

Agenda (tentative)

Robotic Visionary System

Robotic Auditory System

Spoken Dialogue System

Other Sensors / Applications

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Robotic Visionary System

Robotic Visionary System

- Camera on robot

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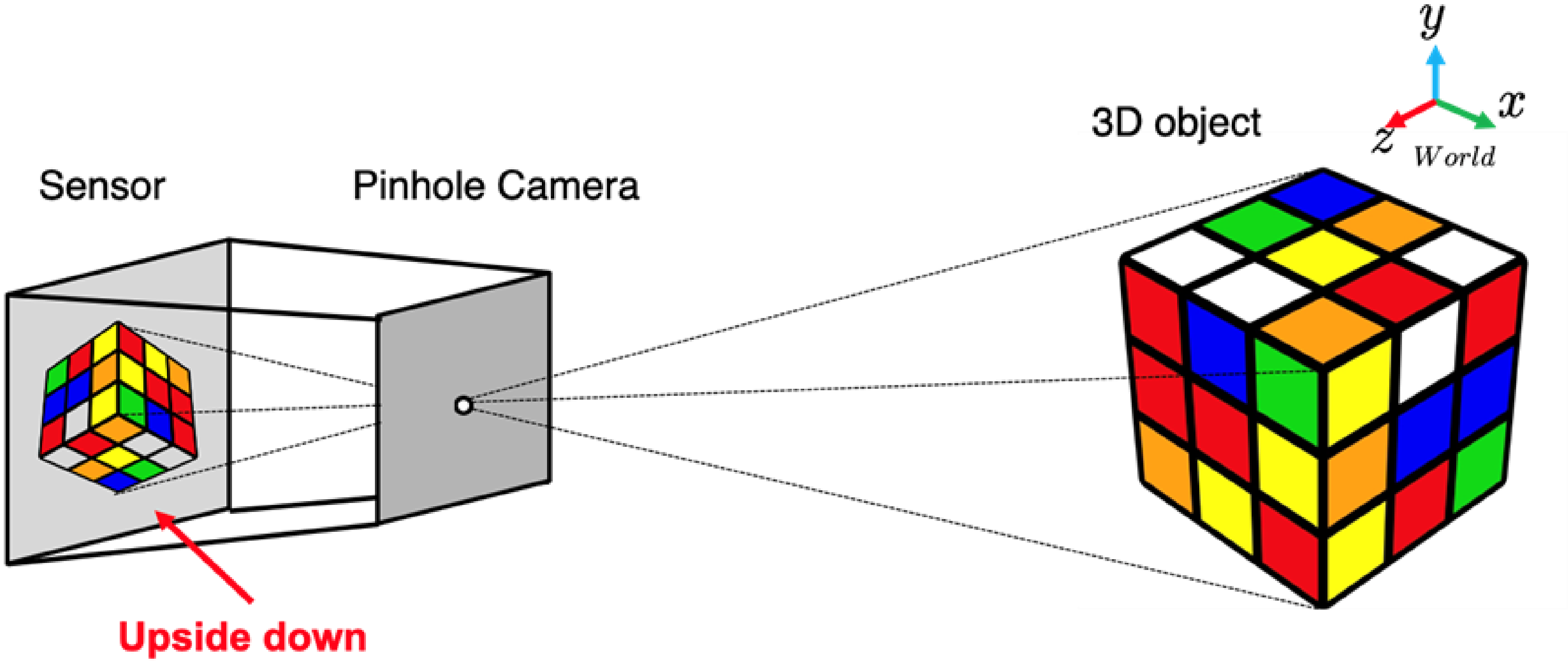
Robotic Visionary System

- Image Formation and Sensing



Robotic Visionary System

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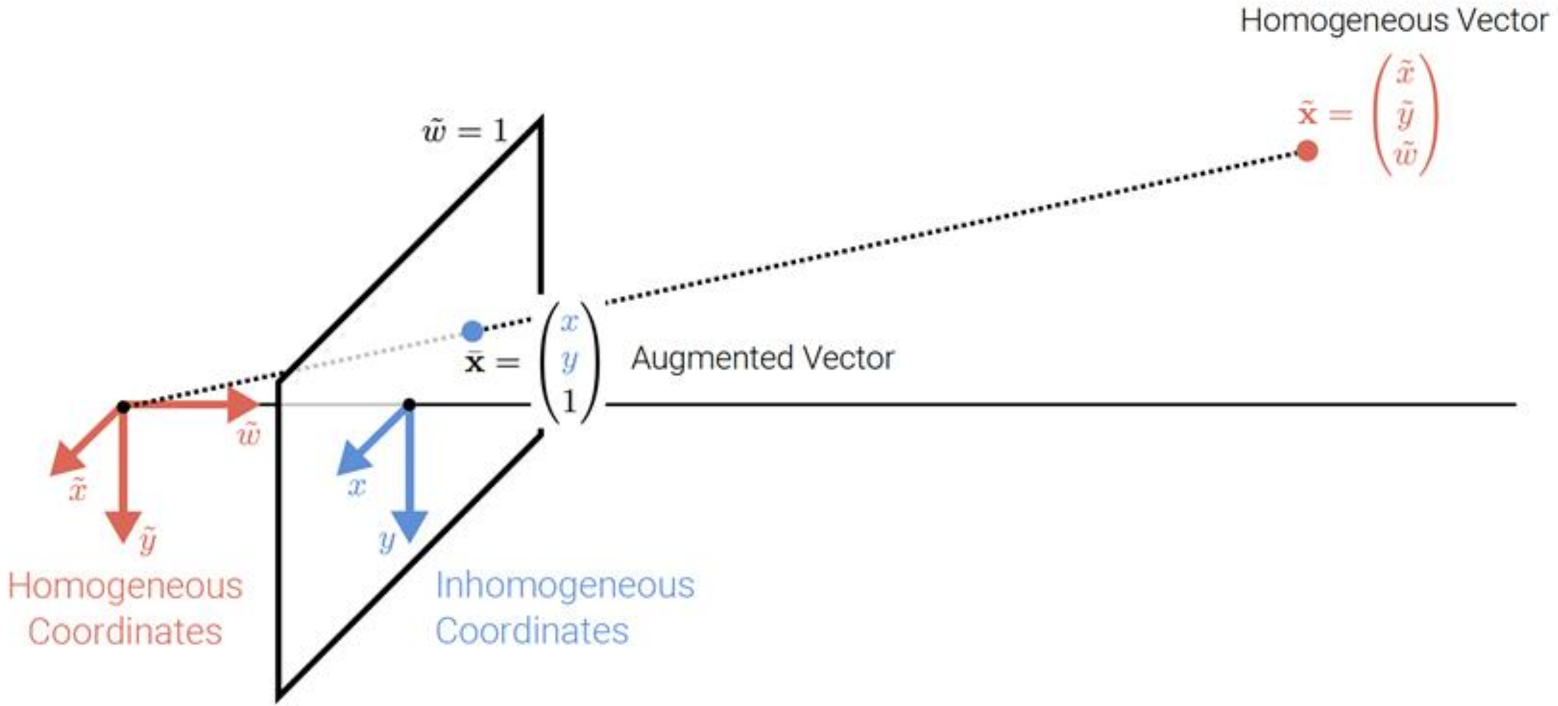
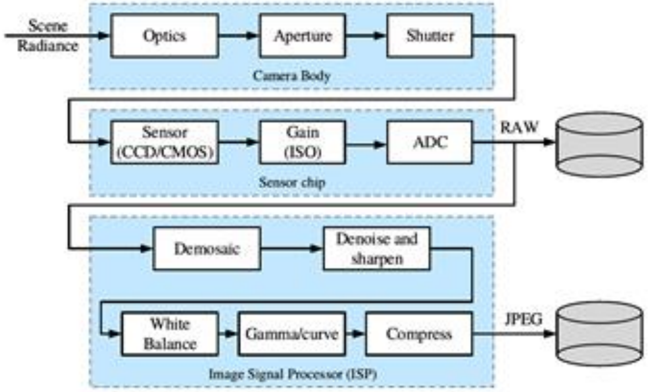


Robotic Visionary System

- Image Formation and Sensing

Image Sensing Pipeline



The **image sensing pipeline** can be divided into three stages:

- **Physical light transport** in the camera lens/body
- **Photon measurement** and conversion on the sensor chip
- **Image signal processing (ISP)** and image compression

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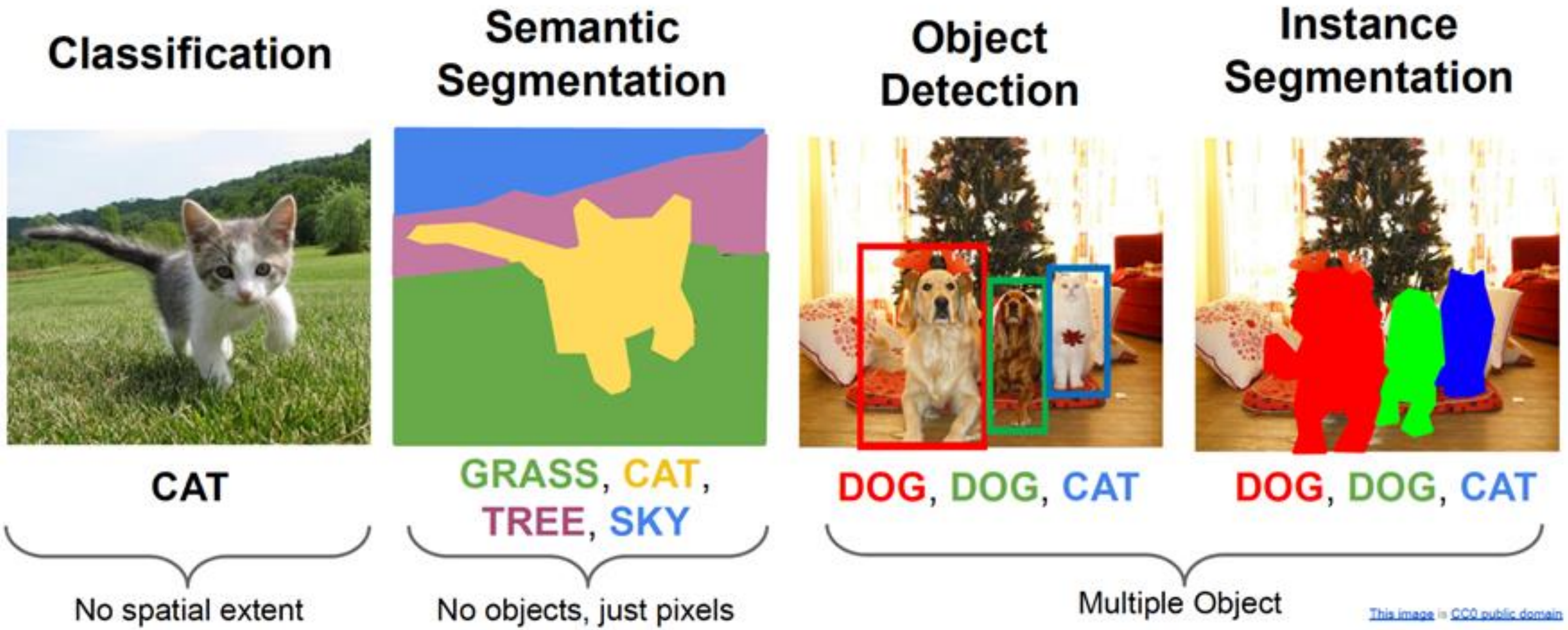
Robotic Visionary System

- Image Processing / Computer Vision



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Robotic Auditory System

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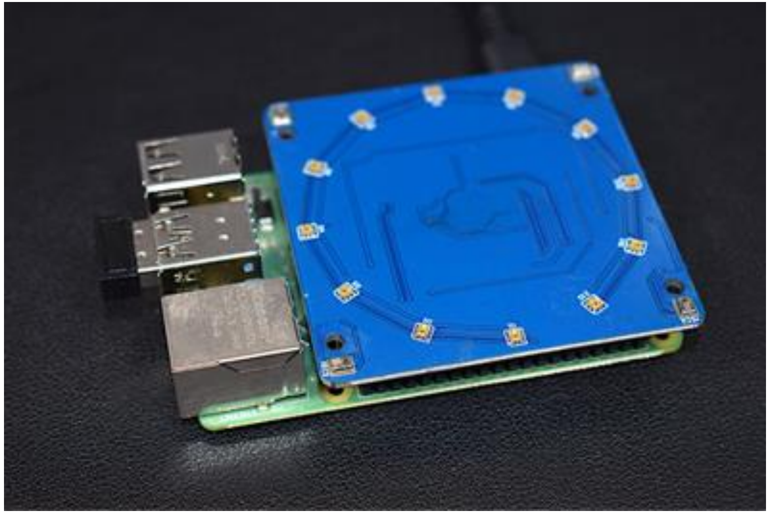
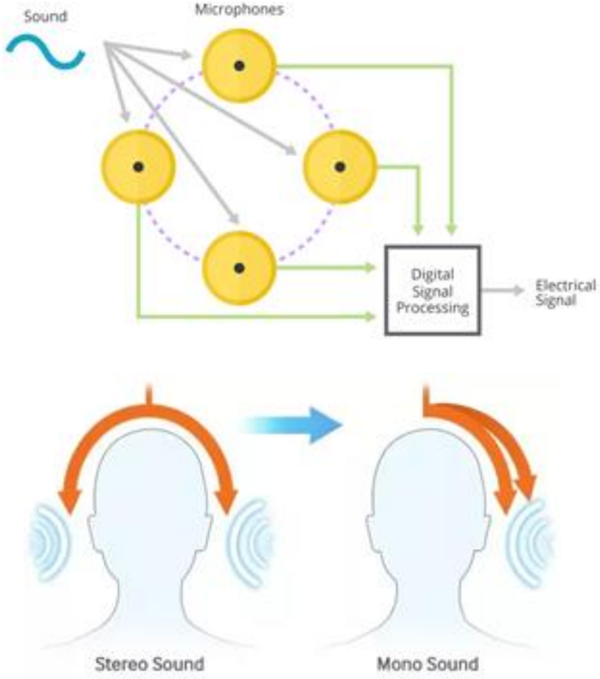
Robotic Auditory System

- Sound and speech acquisition

Robotic Auditory System

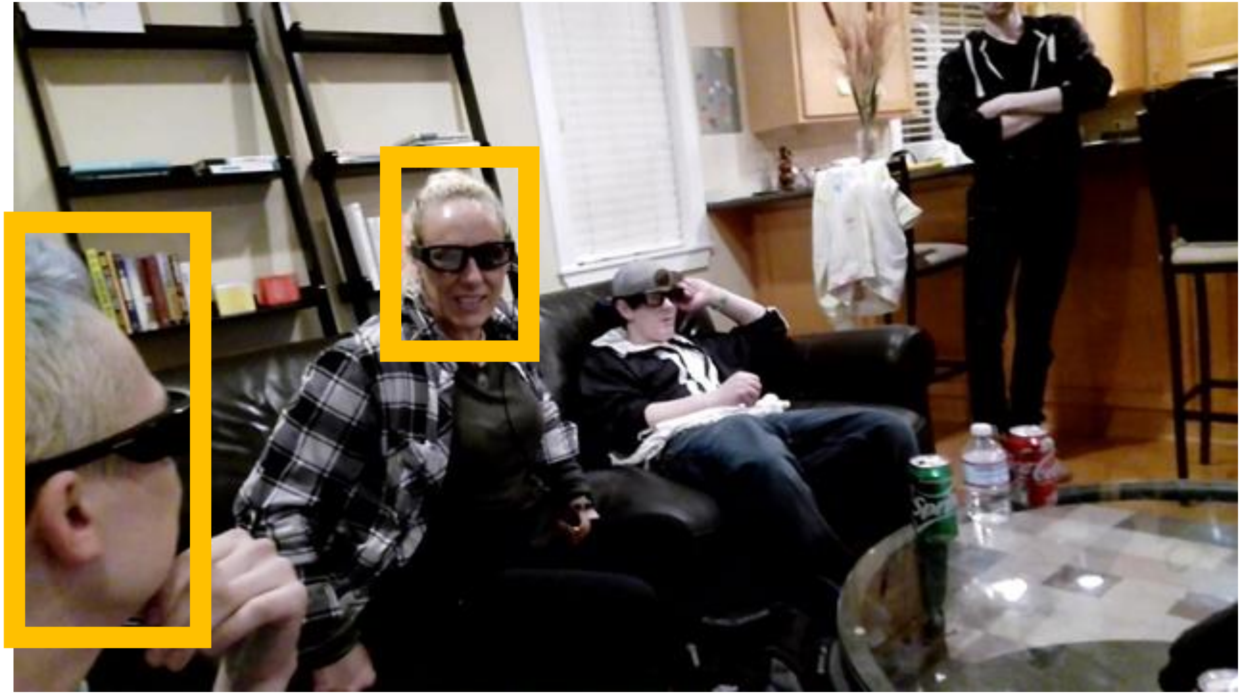
- Microphone array: A microphone device that functions just like a regular microphone, but instead of having only one microphone to record sound input, it has multiple microphones (2 or more) to record sound.





Robotic Auditory System

- Localize sound in space (Sound Source Localization)



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Robotic Spoken Dialogue System

- Given the transcripts (i.e. Spoken Dialogue) → Natural Language Understanding, Conservational AI

Task-Oriented Systems

Users and systems have aligned goals.

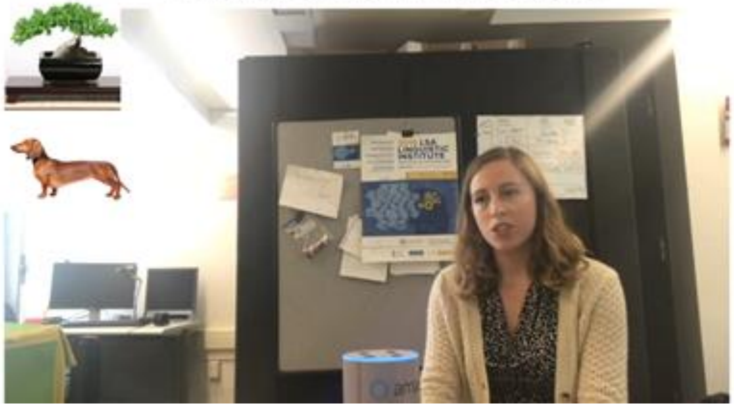
Flight booking

Restaurant booking

Bus Information

Social Chatbots

Talk about anything with anyone.



Robotic Spoken Dialogue System

- ChatGPT

Multimodal Large Language Model

GPT-4 visual input example, Extreme Ironing:

User What is unusual about this image?

Source: `https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg`

GPT-4 The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

Other Robotic Sensors

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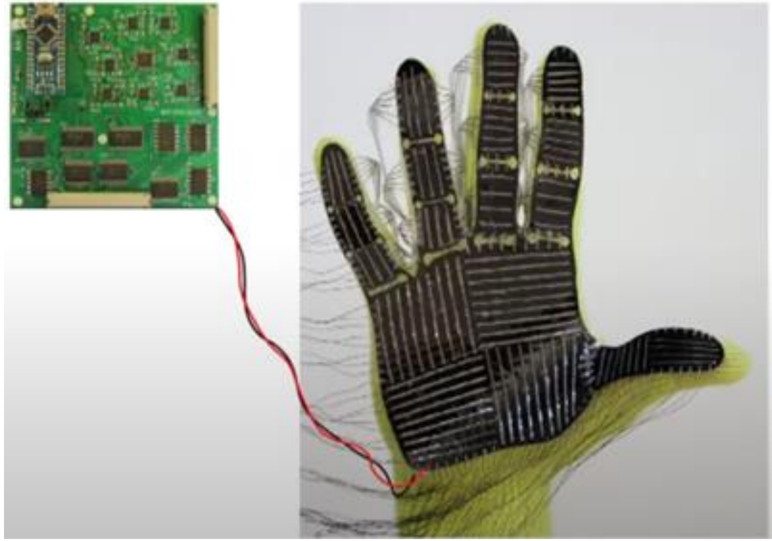
11 Sep, Fri, 4:30-5:30pm	TA office hour (hybrid: outside ECE dept office + zoom link) -- Yanzhe	
15 Sep, Tue, 12-2pm	Object detection, Object segmentation	
18 Sep, Fri, 4:30-5:30pm	TA office hour (hybrid: outside ECE dept office + zoom link) -- Guian	Lab Project Session: TAs will coordinate, some practice exercises for the actual lab project (RCNN inference)- not graded
Recess Week		
29 Sept, Tue, 12-4pm	(In-person) Graded Lab Project: object detection -- Location E1-06-03	
29 Sept, Tue 4pm	48h Assignment 2, Tue 4pm to Thu 4pm -- Yanzhe	
02 Oct, Fri, 10-11am	Robotic auditory system	
02 Oct, Fri, 4:30-5:30pm	TA office hour (hybrid: outside ECE dept office + zoom link) -- Yanzhe	Explain Assignment 2
06 Oct, Tue, 12-2pm	Spoken dialogue system	
09 Oct, Fri, 4:30-5:30pm	TA office hour (hybrid: outside ECE dept office + zoom link) -- Guian	
13 Oct, Tue, 12-2pm	Spoken dialogue system	
16 Oct, Fri, 10-11am	Spoken dialogue system	
16 Oct, Fri, 4:30-5:30pm	TA office hour (hybrid: outside ECE dept office + zoom link) -- Guian	
20 Oct, Tue, 12-2pm	Spoken dialogue system	
23 Oct, Fri, 4:30-5:30pm	TA office hour (hybrid: outside ECE dept office + zoom link) -- Guian	
27 Oct, Tue, 12-2pm	Range and Tactile sensors	
30 Oct, Fri, 10-11am	Lab Project Feedback	
30 Oct, Fri, 4:30-5:30pm	TA office hour (hybrid: outside ECE dept office + zoom link) -- Guian	
03 Nov, Tue, 12-2pm	Application: Video, Video-Language-Action	
03 Nov, Tue 4pm	48h Assignment 3, Tue 4pm to Thursday 4pm -- Yanzhe	
06 Nov, Fri, 4:30-5:30pm	TA office hour (hybrid: outside ECE dept office + zoom link) -- Yanzhe	Explain Assignment 3
10 Nov, Tue, 12-2pm	Application: Video-Language-Action, World Model	
13 Nov, Fri, 10-11am	Exam Review	
13 Nov, Fri, 4:30-5:30pm	TA office hour (hybrid: outside ECE dept office + zoom link) -- Yanzhe	

Other Sensors

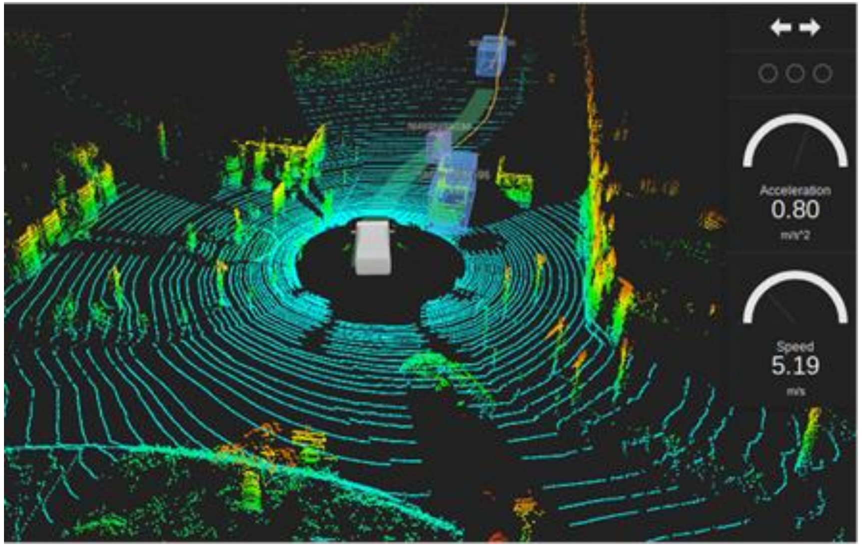
Tactile Sensors



Credit to Wojciech Matusik

Range Sensors

- LiDAR: Light Detection And Ranging.
- Determines ranges (variable distance) by targeting an object with a laser and measuring the time for the reflected light to return to the receiver.



Applications

Agenda (tentative)

11 Aug, Tue, 12-2pm	Introduction	
18 Aug, Tue, 12-2pm	Poses, transformations, image formation	
21 Aug, Fri, 4:30-5:30pm	TA office hour (hybrid: outside ECE dept office + zoom link) -- Guian	Tutorial (Python)
25 Aug, Tue, 12-2pm	Poses, transformations, image formation and sensing	
25 Aug, Tue	48h Assignment 1, Tue 4pm to Thursday 4pm -- Yanzhe	
28 Aug, Fri, 10-11am	Review deep learning (Part I)	
28 Aug, Fri, 4:30-5:30pm	TA office hour (hybrid: outside ECE dept office + zoom link) -- Yanzhe	Explain Assignment 1
01 Sept, Tue, 12-2pm	Image sensing live demo; Review deep learning (Part I)	
04 Sept, Fri, 4:30-5:30pm	TA office hour (hybrid: outside ECE dept office + zoom link) -- Guian	Tutorial (PyTorch)
08 Sept, Tue, 12-2pm	Review deep learning (Part II)	
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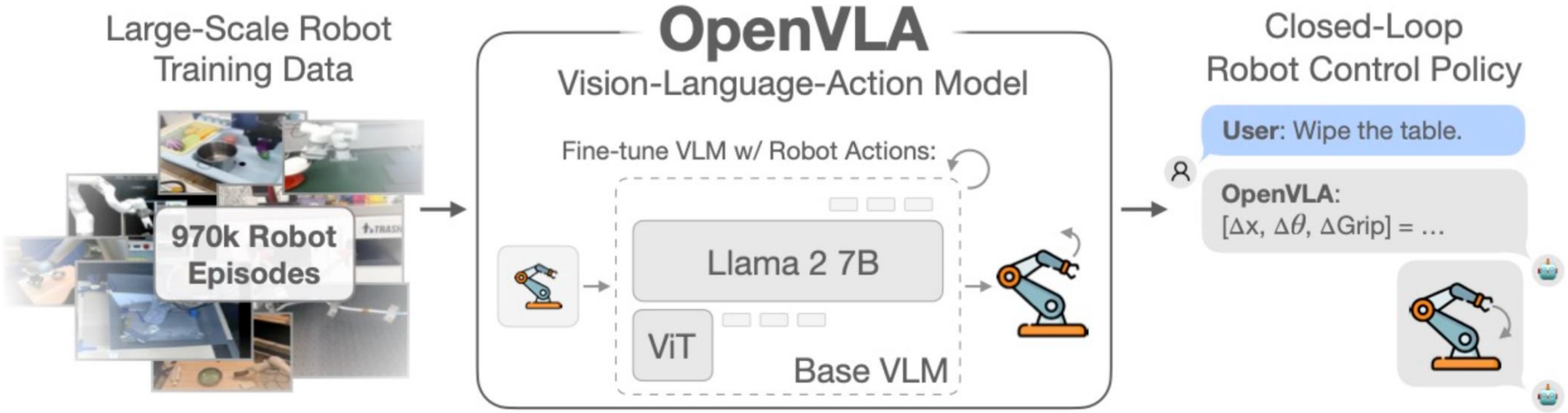
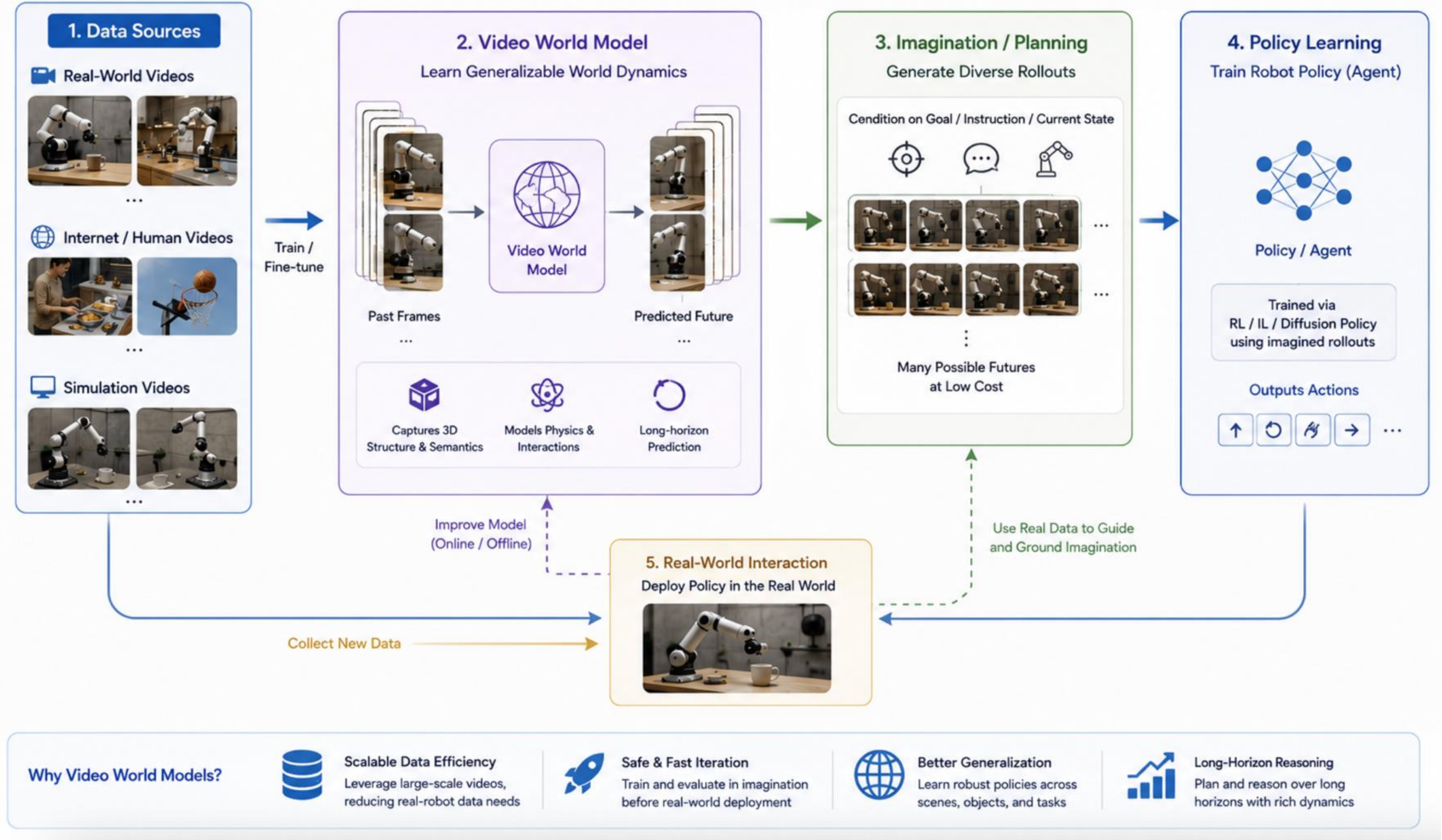
Applications

Video-Language-Action



World Model


1. Data Sources
Real-World Videos
Internet / Human Videos
Simulation Videos
Train / Fine-tune
2. Video World Model
Learn Generalizable World Dynamics
Past Frames
Video World Model
Predicted Future
Captures 3D Structure & Semantics
Models Physics & Interactions
Long-horizon Prediction
3. Imagination / Planning
Generate Diverse Rollouts
Condition on Goal / Instruction / Current State
Many Possible Futures at Low Cost
4. Policy Learning
Train Robot Policy (Agent)
Policy / Agent
Trained via RL / IL / Diffusion Policy using imagined rollouts
Outputs Actions
Improve Model (Online / Offline)
Use Real Data to Guide and Ground Imagination
5. Real-World Interaction
Deploy Policy in the Real World
Collect New Data
Why Video World Models?
Scalable Data Efficiency
Leverage large-scale videos, reducing real-robot data needs
Safe & Fast Iteration
Train and evaluate in imagination before real-world deployment
Better Generalization
Learn robust policies across scenes, objects, and tasks
Long-Horizon Reasoning
Plan and reason over long horizons with rich dynamics

Take a break...

Lecture

- Start time and end time
    a. Tue, Noon-2pm + Fri 10-11am (even weeks, starting from 3rd week)
    b. Something I learned when I joined NUS -- need to end lecture 15 mins for 1h lecture & 25mins for 2h lecture before actual end time so that to allow for students transition time between classes

Lecture

- Teaching Style
  a. 15-20 mins, then stop
  b. 5-mins QA, please directly raise hand and ask, or post question in zoom
  c. Continue and repeat

Assessment

- Take-home assignments (20%)
  a. 3 open-book assignments, to be finished within ~~24~~ 48 hours after release.

- Project (20%)
  a. Basic coding exercises for Python and PyTorch
  b. Implement a robotic visionary system for object detection

- Final Exam (60%)
  a. In-person
  b. one A4 sized help sheet

Timetable

- Website (checkout latest lecture schedule): https://sites.google.com/view/nus-ee4309-2627/home

- Lectures
  a. Weekly on Tue 12-2pm
  b. Biweekly on Fri 10-11am (even weeks, starting from 3rd week)

- TA office hours

Find out more information

- canvas
    a. Find course materials
    b. Find TA zoom conference link
    c. Post Queries in Discussion section instead of email, so that your fellow students can check for similar questions (unless personal query)
    d. Download & upload assignment & project submission
    e. Make announcements

References

- https://shenlong.web.illinois.edu/teaching/cs598fall21/lectures/
- https://uni-tuebingen.de/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/computer-vision/
- https://cs231n.github.io/

Survey:

Q1

1. Given matrices $A = \begin{pmatrix} 3 & 2 \\ -1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 5 \\ -2 & 4 \end{pmatrix}$, what is the product AB?

A. $\begin{pmatrix} 3 & 10 \\ 2 & 0 \end{pmatrix}$

B. $\begin{pmatrix} -1 & 23 \\ -1 & -5 \end{pmatrix}$

C. $\begin{pmatrix} 4 & 7 \\ -3 & 4 \end{pmatrix}$

D. $\begin{pmatrix} -1 & -5 \\ -1 & 23 \end{pmatrix}$

Q2

2. Which of the following statements about NumPy array indexing is **true**?

You can use a boolean array of the same shape as a NumPy array to filter and select elements from it.

Slicing a NumPy array returns a new, independent copy of the data, so changes to the slice won't affect the original array.

Q3

1. Given the input array $X = [3, 1, 2, 4, 0]$ and a 1-D convolutional filter $W = [1, 2, 1]$, what is the output array after applying the filter with a stride of 1 and no padding?

A. $[3, 2, 2, 4, 0]$

B. $[6, 8, 4]$

C. $[4, 5, 6]$

D. $[8, 7, 10]$

Q4

4. In the self-attention mechanism, the **Query (Q), Key (K), and Value (V)** vectors are derived from which input?

A. Three different tokens selected from random positions in the input sequence.

B. The Positional Encoding vectors.

C. The same input token embedding, which is then projected into three different vectors.

D. The output of the previous layer's feed-forward network.

Thank you!